Sourav

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Education

Graphic Era Hill University (GEHU) 2024 – 2028

B.Tech in Computer Science and Engineering

Technical Skills

Languages	Java, C++, Python, TypeScript, Kotlin, SQL, Bash
Backend	FastAPI, Flask, REST APIs, JWT, SQLAlchemy, PostgreSQL, SQLite
Frontend	React 19, TypeScript, Vite, Material UI, Tailwind CSS
Systems	Linux (Fedora), TCP/IP, raw sockets, libpcap, POSIX APIs, multi-threaded programming
Security	ECDH key exchange, AES-256-GCM, end-to-end encryption, JWT authentication
Tools	Git, GitHub Actions, CMake, Maven, Playwright, GDB, Valgrind
Concepts	Data Structures & Algorithms, Operating Systems, Computer Networks, OOP

Projects

DPI Engine — Deep Packet Inspection System *C++17, CMake, libpcap, Linux, GitHub Actions*

- Engineered a multi-threaded network traffic inspection engine with a producer-consumer pipeline, using consistent hashing over 5-tuples to route packets to load-balanced fast-path workers with per-flow connection state tracking.
- Parsed Ethernet, IPv4, TCP, and UDP headers; extracted TLS SNI and HTTP Host fields for application classification across 20+ signatures, with rule-based blocking by IP, domain (wildcard), application, and port.
- Established CI via GitHub Actions with GCC/Clang/MSVC matrix builds across Linux/macOS/Windows, CTest unit tests, clang-tidy static analysis, and clang-format enforcement.

Khoji — Offline AI Knowledge Workspace *TypeScript, Python, Rust (Tauri v2), FAISS, Playwright, SQLite*

- Built a cross-platform desktop application using Tauri v2 (Rust) that ingests PDFs, DOCX, PPTX, EPUB, and images; extracts text, builds semantic search via all-MiniLM-L6-v2 embeddings + FAISS vector index, generates flashcards and quizzes, and runs a RAG-powered LLM chat (Qwen GGUF) – all 100% offline.
- Architected Python AI engine with multi-stage extraction pipeline; implemented SQLite persistence (WAL mode, cascade deletes) for document metadata, embeddings, and generation cache; shipped as AppImage with 32 commits and a tagged production release.
- Wrote Playwright E2E tests across Chromium and Firefox; integrated automated verification suite (ruff lint, IPC smoke tests, engine E2E) triggered on every backend change.

PeerLink — Serverless Secure P2P File Transfer *Java 21, JavaFX, TCP/IP, ECDH, AES-256-GCM, Maven*

- Built a serverless peer-to-peer file transfer application over direct TCP socket connections, eliminating single-point-of-failure without a central relay or cloud dependency.
- Implemented end-to-end encryption using ECDH (secp256r1) session key agreement and AES-256-GCM authenticated encryption, ensuring confidentiality and integrity against MITM attacks.
- Leveraged Java 21 Virtual Threads and a BlockingQueue pipeline for concurrent multi-file transfers, decoupling encryption from I/O for throughput efficiency; packaged as native installers (AppImage, DEB, RPM, MSI).

PeerLink Android — Secure P2P File Sharing *Kotlin, Android SDK, TCP/IP*

- Developed an Android companion to PeerLink enabling direct device-to-device file transfers over local networks with the same zero-trust security model, using Kotlin coroutines for async I/O.

Leadership & Activities

- **Technical Club**, GEHU — co-delivered a workshop on transformer architecture and LLMs for 100+ attendees.
- Co-organized a 200+ participant university hackathon: managed technical infrastructure, participant onboarding, and multi-track event logistics.
- Daily Linux (Fedora) user; proficient with command-line tools, debugging (GDB, Valgrind), and build systems (CMake, Maven).